ActInf GuestStream #039.1 ~ Sources of Richness and Ineffability for Phenomenally Conscious States

Source: YouTube **Playlist**: GuestStreams **Duration**: 1:54:41 | **Uploaded**: Unknown **Transcript method**: auto_caption

---

hello and welcome this is active guest stream number 39.1 it's March 16 2023 we are here with shuji Eric elmoz Nino and Guillaume Dumas we're going to have a presentation followed by a discussion section so thank you all for joining and off to you for the presentation all right thanks Daniel uh yeah I'm Eric I'm a PhD student in uh Joshua bengio's lab I'm shuji I'm a postdoc with Joshua yeah no we're really excited to be here thanks for the invite um we're gonna be talking about why we can't describe conscious experiences um so this is this is going to be our take on on a really long-standing problem in the philosophy of mind but we're going to be looking at it through the lens of computational Neuroscience and information Theory so uh yeah hopefully it'll be fun for everyone it mixes a bunch of disciplines so I think the the most Salient way to illustrate the problem that we're going to be uh addressing is to ask you to try and think about how you would describe the experience of seeing the color red um so probably the kind of stuff that's going in your head is things like well it's a bright aggressive color you know symbolizes love that sort of stuff and um in a sense this this is the description um it's effective I would be able to to guess which color you were describing uh but in a in another sense it's really inadequate right if I were blind for instance um your description would be totally useless I would I would be no better in understanding what red looks like so there's this real sense in which conscious experiences are ineffable such that we can't describe them and this applies to percepts like red but it also applies to experiences more broadly like right the the experience of you know having a thought is just so ineffable so hard to describe it um and really importantly this this doesn't happen with most of our knowledge all right I could describe most of what I know except for experiences they have this special place so uh it's a big Topic in the philosophy of mind um because it relates a lot to this thing called the the hard problem of Consciousness so basically all the hard problem is it's uh it's the problem of how and why physical processes give rise to conscious experiences in the first place um so this is probably the the oldest and most debated problem in the philosophy of mine and it's even led many to the conclusion that you know actually we're going to give up and and Consciousness can't be explained with physical theories at all you know that it it can't simply emerge from Neuroscience computation or known physical laws um so that's that's a bold statement and I want to get into all the details uh about what makes the heart problem so Salient uh but just here's a few here's a few things um so first of all um we could we could logically conceive of what are called philosophical zombies uh philosophical zombies uh very different from Hollywood zombies they're basically beings that are physically identical to us they behave exactly like us in fact they even have the exact same neural activity that we do um but they just aren't conscious all right so similarly instead of these these alternates uh agents being unconscious we could also Imagine them as just experiencing different things when they're in the same States uh so for instance it's conceivable at least that in an alternate universe the same exact brain mechanisms that produce an experience of red in this world would instead produce an experience of green and the other and that all experiences would kind of be like flipped in this way um so you know intuitively we think that these these two thought experiments aren't actually possible in our universe but even just their logical conceivability suggests some kind of explanatory Gap where Consciousness doesn't seem to neatly emerge from or be determined by you know physical state or function um another really prominent thought experiment is uh the knowledge argument and this is the main sort of problem that our work is going to address we're not going to address all facets of the the hard problem of Consciousness but um this knowledge argument is one of the uh the biggest arguments against this against fiscalism so how the knowledge argument goes is uh it's another thought experiment it asks us to imagine um Mary who has grown up in a black and white room her whole life uh and you know this despite this poverty and stimulus she actually knows uh a lot in fact we're going to assume that she knows all the physical facts about how color perception uh works so all the relevant physics all the relevant Neuroscience Etc uh and now we're going to ask well does she learn something new when she steps out of the room for the first time and actually experiences color right and if we say that she does learn something new which which I think is the intuitive answer there seems to be a problem right we assume that she knew all the relevant physical facts about color perception so doesn't this mean that what she learned had to be something that was non-physical um and if there was nothing more to color perception than you know physically embodied information and neural activity then the experience is something describable uh that she presumably could have read about understood and would have already known so it feels like this this thought experiment really pushes us against uh physicalism to towards this idea that maybe you know conscious experience is has something on top of just information content um now of course we don't want to reject physicalism it's been been very fruitful for us in the history of science uh so we really do want to say that experience just consists of information that can be described so there has to be something wrong in this uh this knowledge argument here and what we're going to do is we're going to bite the bullet and acknowledge that experiences actually are ineffable so maybe it is the case that experiences can be described in principle since they consist of nothing more than information um but maybe they can't be described uh in practice or at least they can't be fully described in practice using something like language so then the challenge becomes explaining why experiences are ineffable under a physicalist framework and that's what our work is uh is really going to be about okay so the story the structure of the talk is going to be like this summarize briefly why uh characterizing richness and ineffability is an interesting and important question then Eric is going to talk about how there's a natural correspondence between ineffability and information loss which is going to let us link biologically plausible attractor models of working memory to ineffability essentially because information loss is inherent in attractive Dynamics then I will talk about um if you consider the ineffability of conscious experience to verbal report as just a special case of information lost between two specific points in the communication pipeline then you can actually generalize the notion of ineffability more broadly by considering for example loss from sensory processing to conscious experience and even interpersonal information loss um we're going to prove using colgoma of mutual information that your conscious experience being ineffable to another person implies High cognitive dissimilarity between the two of you um under our model and finally we'll talk about um whether an exact definition of conscious experience is required at least for characterizing the nature of ineffability and richness and also some open questions uh so yeah why why is richness and ineffability an important question um there are sort of several reasons so first as as Eric um said um understanding Consciousness is not just of general interest it's a long-standing central problem in philosophy of mind um because it's not easy to reason about um which has led some Duelists to believe that Consciousness must be at least partly non-physical in nature because they can't see how it could be explained by physical processes um but it's also a topic that is of great interest in machine learning um because if you assume that Consciousness is essential to uh human cognition then it follows that we won't be able to engineer machines that think like humans without incorporating consciousness um and uh so we're actually going to argue in this talk that the confusing aspects of human consciousness such as its ineffability um can be understood with information Theory um which as you all know is born from computer science um we're going to use some Primitives from from information um Theory to shed light on why Consciousness is ineffable and hopefully to dispel uh some of the Mysteries surrounding it all right so uh one of the main Primitives over here that we're going to use is um to to talk about uh attractor Dynamics uh in the brain an unconscious experience and how that results in uh ineffability so hopefully that this sort of perspective is uh familiar to most people but um we could talk about neurodynamics by saying the brain has a state at any particular time and we could denote this state using a vector so for instance um in this plot over here you know maybe we could denote each axis in the state each Dimension uh using an individual neuron and maybe uh we denote the the activity of each neuron using its average firing rate right so any states at a particular time would be the firing rate of all the neurons across the brain um now there's nothing special about neurons or firing rates you know if you think that things like the synapse strengths or the dendrite potentials or the astrocytes or whatever else is also important to the representation you could include those in the state they're just additional axes um the details aren't so important as long as you know we have some state that describes uh the what the brain is doing at any particular time um now of course the brain is a dynamical system these states are changing uh so you can think of of craping out trajectories through State space as a function of time um and the way that uh the the state evolves will be determined just by the recurrent connections within the brain as well as inputs coming from uh elsewhere maybe we're just talking about the Dynamics within a given brain region the inputs would be you know the other brain regions talking to it um as well as any sensory stimulus coming in right so this dynamical systems perspective of the brain it's been really instrumental in understanding the computations it performs and one of the primary methods for characterizing these computations is to look at what are called uh the state attractors of the system and for us as well state attractors are going to be a really important part of our ineffability framework in a bit um so basically what a tractor state is is as a states where once the system reaches it it's going to remain there at least until inputs come in and change the Dynamics or noise not just the state out of this attractor um so you could see this thing clearly within the figure so so the the X and Y axis over here are like the state of the brain again maybe something like the average firing rates of the neurons these arrows are illustrating the the Dynamics of the brain so how the state will evolve given the brain's connectivity given the synapses and you can see that there's some regions where all the arrows all the Dynamics kind of contract to a single point uh so that's called a fixed points if it's one-dimensional it's a kind of state attractor and because all the arrows Point towards it you know once you finally reach that state you're not going to move out again until the Dynamics change because inputs came in or until noise nudges you out and you can see that each each state attractor also is associated with this this sort of attractive region an attractor Basin it's called where you know once a trajectory is somewhere in that region it will eventually converge to the attractor and in the absence of noise um so these These attractors are really common in the brain uh and also in uh artificial recurrent neural networks uh trained on tasks and a big computational benefit is that they provide a form of transient memory right once you're in the attractor you remain there that's for where the memory comes in uh and because of this and other benefits they're implicated in a wide variety of mineral processes like working memory long-term memory decision making um and higher level cognition and we'll argue Consciousness uh in in general so they're they're all over the place uh in the brain um an additional aspect of them that's uh interesting and that's going to be relevant for our talk is that you can think of attractors as having a sort of dual discrete and continuous nature um so what I mean by that is that the discrete aspect of an attractor is given by the fact that there's a finite number of them and that attractors are mutually exclusive right you're in one or you're in another so this means that we could use discrete symbols to denote which attractor the system is in right there's a finite number of them you're either in one or another and this basically lets us you know identify or label the states using a discrete variable um there's also a notion though in which an attractor is just a continuous High dimensional Vector in this this really high dimensional State space of the brain right every attractor basically is just a vector in state space and that means that they're just not they're not arbitrarily discrete States they have continuous representations we can say for instance that you know one attractor is more or less similar to another one depending on how close they are in state space um and and that that gives a representation to the attractor says that the high dimensional sort of continuous part so just to recap the screen part is you could you could identify which attractor you're in in the finite set of all possible attractors um but you could also uh talk about an attractor through this like continuous High dimensional Vector that describes where the attractor is in the high dimensional State space of the brain um and you know to wrap your head around this there's a a good analogy to word embeddings in NLP so in NLP you know each word has a discrete ID that's like the symbolic discrete part but also each word is associated to a high dimensional Vector that gives it a representation sort of the meaning of the word where we could say that you know words are more or less similar uh to each other okay so we're going to use attractor Dynamics quite a bit to explain a significant source of ineffability in conscious experience later uh but for that to be relevant I first need to spend a bit of time to convince you that neurodynamics that produce conscious experience uh actually do have attractors um and for this there's already a lot of work in the literature connecting attractors to conscious experience so this part is not something uh new from from our work so for instance um one of the leading theories of Consciousness which is the global workspace Theory actually implicitly implies uh that attractor Dynamics are kind of fundamental uh so what Global workspace Theory says is basically like across the brain and all the different regions you have sort of uh sub-process worker workers uh basically and these different distributed mechanisms they can't speak to each other uh very easily they mainly communicate through this like Global workspace it's kind of like a bottleneck I think of it like a Blackboard uh that all the different processes across the brain could write to you could also think of it as like the ram of the brain but basically it's collecting information from the different workers um and one key uh aspect about uh the the global workspace it's posited and the theory is that for content in the workspace to then be broadcast to the to the sub processes for instance for us to be able to um you know talk about what's in the workspace for for it to be written to uh kind of speech regions the content of the workspace has to be Amplified and sustained for a certain amount of time now space stable basically and this is clearly where the uh the attractors come in right attractors by definition are these states that are that are stable these states that are sustained uh so basically what Global workspace Theory says is that the context of the global workspace um that can be used by the rest of the brain uh are are the attractor States and the relevance to conscious uh to Consciousness in this theory is that um it posits that the contents of This Global workspace are you know the the things that we're aware of the things that we're able to report the things basically that we're conscious of um and then just just kind of like terminologically another term for this workspace is called uh working memory um the idea there is it's a sort of like really short-term memory in the sense that you could quickly um you know act on the things that are in this Global workspace or report on them they're accessible basically uh okay so so that's that's one very powerful argument uh for why attractors are relevant to Consciousness there's a bunch of other related arguments uh for instance just introspectively you could think of um you know what experience is like is you know a sequence going from one thought to the next to the next we kind of have this like a sequence of discrete events um and that that clearly looks like a dynamical system kind of jumping between uh attractor States uh and then experimentally uh there's also been work showing that in in Psychology experiments when you sort of uh present the stimulus to a subject and you vary the amount of time uh that the stimulus is presented to kind of bring it below or above the the conscious threshold well in the conditions were subject to report being aware of the stimulus the main thing that's that's different is that the representation is really stable and robust in noise and those are also clearly properties of uh of attractors okay so uh again like we'll use those attractors to identify a source of ineffability uh in a little bit but first I need to also introduce how we're going to formalize the the very notion of uh ineffability and for this we're going to use information Theory um part Shannon information Theory I'll do that now and then uh she will talk a bit about comograph complexity as another formalism uh but uh hopefully most people are are familiar with this um but one thing that we're going to be using is the entropy of a state uh so for instance if you have the entropy of some random variable X to make things concrete maybe that random variable is conscious experience so the entropy of conscious experiences um that that value would be high if the distribution over conscious experiences was kind of a diffuse and flat so there's equal probability for all possible experiences and entropy would be low if the distribution over conscious experiences was peaky so that's kind of like the the green distribution in this image would have low entropy and the red distribution that's sort of flatter and more uncertain would have higher entropy and um basically what what entropy is going to quantify in our framework is the richness of a variable in particular the richness of conscious experiences why this is a good measure of richness is because if conscious experiences could kind of take on many possible values with people probability they're very kind of diverse in that sense well then it's it's something that's rich um some other Notions that are going to be important are mutual information uh so here that's denoted by um eye of actin Y and what that denotes is how much shared information there is between two random variables uh so again to make things concrete X we could take to be a conscious experience whereas why maybe that's a description of the experience maybe a verbal description so in this case the mutual information will be really high if the verbal description uniquely determines the um the experience so basically another way of saying that is knowing the verbal description we have no more uncertainty about what experience is going on they perfectly determine each other so that would maximize the mutual information and in contrast If the message told you nothing about the conscious experience and you were just as unsure as when you didn't have the message well then the mutual information would be zero so this is this is uh useful because we could use it to define a notion of um so this this conditional uh entropy that we've written h of x given y uh is equal basically to the richness of X so the the entropy of a conscious of conscious experiences uh minus the mutual information between conscious experiences and uh other variables like their descriptions that's the mutual information what this captures basically is how much information is lost when going from an experience X to a description y right if all the information was lost if y doesn't tell you anything about X the mutual information is zero and H of X will uh well the conditional entropy will be equal to to the richness of x if y perfectly determined X well then h of X and the mutual information cancel and um the conditional entropy would be zero so what conditional entropy uh describes basically is how much information is lost when moving from X so like a conscious experience to why a description of it and that's that's a perfect uh description of what nfability is intuitively right it's how much information is is lost when I try and describe something like an experience okay so uh I think we have we have uh most of the uh the background of the way and now we're ready to uh describe one main source of ineffability that uh arises due to attractor Dynamics so very quickly to describe some some variables over here that we're going to be using throughout the presentation if you look at that top the top right diagram we're going to have X which is basically the trajectory of neural activity that's relevant for a conscious experience so for instance in global workspace X would be the trajectory of the working memory state uh we're going to assume that that this uh this this trajectory of neural activity produces a conscious experience we're going to be calling that conscious experience s throughout but also the trajectory importantly is going to follow attractor Dynamics right so they converge the states such as at so how does this result in ineffability well basically the idea is that whenever you have a tractor Dynamics there's a many-to-one mapping from trajectories to attractors and this induces information loss right just knowing the attractor we're not able to kind of uh go go the other way around and and infer what the original trajectory was so that's there's this conditional entropy this ineffability of x given a is going to be high basically all right why does this matter who cares if there's information lost between the trajectory of working memory and the attractors that it converges to uh well important thing here again is that Global workspace postulates that four working memory contents to be accessible across the brain it needs to be Amplified and maintained over a sufficient duration so it's thought to be around 100 milliseconds so this means that you know working memory is going through these trajectories X but only a only the attractor can be reported and only the attractor could broadly affect behavior only it could be written to memory only it could be broadcast to these other these other processes so that means that the contents of working memory uh kind of these transient contents in the the trajectory X are inherently fleeting right there's a rich information that was encoded in the moment in the moment of the trajectory but the cat later be recalled or reported because only the attractors can be recalled or reported um and and kind of if we we zoom out and uh go back to our philosophical Notions of ineffability this also explains why it's difficult to sort of catch yourself in a thought right working memory encodes these rich and subtle thoughts through the trajectory but we can never quite pinpoints or report in words what those thoughts consisted of uh were they verbal were they pre-verbal that they consist of visual imagery it's really hard to say exactly and the reasoning would be that well there's just a lot of information that's that's lost going from these trajectories to the attractors that again we could report on or that we could recall or they could just broadly affect Behavior right so that's one huge source of ineffability arising from uh attractor Dynamics in working memory um now these attractor Dynamics also give a lower bounds for ineffability during verbal reports all right so now at the top right I've added another variable uh M that's basically some some verbal message that you could be using to describe your experience and M is a function of the attractors right according to Global workspace only the attractors can be reported on or used across the brain um so this means that um the the the ineffability of an experience given a verbal message has to be at least as great as the ineffability given an attractor because m is a function of the attractor um and and you can't gain information when applying a function you can only destroy it uh but we're actually going to argue that um the ineffability is not just lower bounded by the ineffability of the attractor it's actually uh in practice quite a bit greater than that and the reason is that M this message is a discrete variable right it's language um whereas a the attractor is this really high dimensional snapshot of some cortical States so because of the asymmetry between the richness of these variables there has to be uh information loss and intuitively what's going on here is that you can think of there as being a many-to-one mapping from attractors to messages as well so for instance if uh I say that I saw Fat Cats um right that paints a picture but it's also leaving out significant details about the original attractor which presumably encoded things that you were aware of like the cat's color size suppose the background all these things that that you were aware of and that were encoded in the attractor but you can't really put into a message without it being prohibitively uh charge so again the idea here is that the ineffability of experiences given messages higher than given attractors because the message divides the space of attractors more coarsely it's it's a simpler variable it adds additional information loss uh so one problem is if if language is so coarse and simple why can like uh why can it describe experiences at all um and uh the solution is that while attractors and the message kind of both share this this uh discrete part um right so the message uh because uh language is discrete can be used to index or identify compositional attractors uh there's this comparable richness basically between a message and a discrete part of the attractor which means that in practice the richness of the experience h of s is um is uh typically much greater than the ineffability of the experience given given the message so that that explains why you know we still can communicate somewhat with with language even though it's this really low dimensional simple thing all right so to recap quickly uh what we have over here is the argument says that detractor Dynamics are empirically ubiquitous in in neural activity across brain regions uh they've been proposed as a computational model for working memory um and and prominent models contends that conscious experience is a projection of working memory States uh and one of our key contributions is to connect these theories to argue that attractor models working memory offer an account for the ineffability of conscious experience because of tractor Dynamics induced significant information loss right that's the general argument um there's there's one quick thing to add over here so I've been talking about um you know working memory a lot less about conscious experience so far uh so so let's link the two um the what's going on over here is that there there's many ways to link conscious experience to X to working memory basically uh and and there's kind of two main options so one is we could say that the instantaneous state of working memory is uh always conscious so all the states in between attractors are also consciously experienced and uh the how how attractors would result in um nfability here is that again what you're able to recall what you're able to report is just the attractor not the actual uh experiences in between attractors that you were having but there's another possibility which is to say well maybe only the attractors themselves are experienced however importantly ineffability exists regardless in this case right if only a is conscious a can still encode the fact that there's information lost during processing and working memory during these trajectories information loss is something that is computable uh it could be encoded along some dimensions in the attractor for instance which explains how you know we could be consciously aware that there's information lost during working memory processing okay yeah um so now I'm going to talk about generalizing ineffability um once you view enough ability as information loss from a source variable to a destination variable Then There are a myriad of different Pathways that you can characterize which we're going to split into two groups um so Pathways confined within an individual or intrapersonal communication and Pathways that extend between individuals or interpersonal communication so in the intra case you've already seen conscious experience the working memory trajectory the attractive State and the message um but you can also consider D which is the input data to the system and V which we use to denote the state of processes considered um outside the limitation of working um and uh you can also consider the cognitive parameters of the individual being communicated to so we're going to work with Alice and Bob um and we're going to assume that Bob has a brain which is structurally identical to Alice's um but has different parameters uh Phi tilde instead of Phi and we're going to denote Bob's um cognitive State using the tilde uh so empirical evidence from Neuroscience suggests that the brain is hierarchical in nature and there are many levels of organization many instances of attractor Dynamics across organizational levels and cortical regions um so one example is the inferior temporal cortex um is a sensory processing area that responds discriminatively to novel sensory stimuli whereas the pre prefrontal cortex or PFC appears to be implicated in maintaining the attention modulated representations of working and to a first approximation um in this simple model it's easy to obtain the results that richness of one process constrains the richness of another uh so here we have that richness of working memory trajectories um HX um is is upper bounded by the richness of its inputs and subsequently uh we also have that the richness of conscious experience HS is upper bounded by the richness of conscious experience and the richness of um sub-process States now we're going to zoom out and consider the interpersonal communication case um Shannon entropy has some drawbacks when it comes to characterizing interpersonal ineffability um a major one is that Shannon entropy assumes you have access to Phi um Alice's cognitive parameters uh so note that H Phi s given M for example is the descriptional um uh complexity of us given not just a but also Phi um and since we can't assume in the interpersonal case that Bob has access to Alice's brain parameters we're going to rely on the colgoma framework for information Theory uh if you're unfamiliar with the congomorov complexity the first thing to note is that Congo of complexity or k x is defined on instances um X hence the lower case rather than variables um we try to notice by capitals um KX is roughly defined as the length in bits LZ of the shortest binary program said that Prince X and holds um so there's no explicit dependency on the probability distribution over X unlike in um Shannon information uh but we can introduce PS um by taking an expectation over the clock complexity of States um to uh obtain the congomorov version of Shannon entropy and likewise for ineffability we characterized it before using conditional entropy um in the kogoma framework expected K of x given y um is is sort of the analog to Shannon conditional entropy and that represents the length of the shortest program that Prince x given that you know why or that Y is um accessible to your program uh and it's uh approximately equivalent up to logarithmic terms um to KX minus i x colon Y where I X codon Y is called more of mutual information which can be interpreted as uh the difference in program lengths for printing X depending on if you know why or not roughly speaking um there are a lot of similarities between Shannon information and program information for example um Mutual information in both cases is maximized if x is equal to Y uh in the Shannon case because that means y uniquely determines X and in the kokomorov case because it means that given y um you don't require any extra bits to print X in fact uh if you assume knowledge of the probability distribution p uh then expected more of complexity and Shannon entropy are equivalent up to some constant Factor because it turns out that the most efficient way to describe a state X on average given that you know the distribution um is to encode x with a a description of length minus log p x bits um but if you don't know the distribution um which we want to make use of uh in in the interpersonal case because the parameters of the speaker's brain are not given um then the higher the descriptive complexity of that unknown distribution then the higher the ceiling on the difference between expected colgoma of complexity and Shannon entropy um uh and in our case we'd expect that to apply because the space of cognitive States is enormous the probability distribution over those States is very complex and the states themselves are in general um complex to reconstruct um being very high dimensional vectors um the second advantage of coconut complexity um in addition to allowing us to explicitly avoid conditioning on the probability distribution is that Shannon entropy is a measure of statistical determinability of States as opposed to difference in unique states um so in Shannon information um uh entropy is fully determined by the probability distribution on States and it's unrelated to the meaning or structure or content of the states um whereas complexity is concerned with the difficulty of reconstructing um uh States I.E absolute difference which corresponds more closely to the lay definition of ineffability um here are a number of results that that we include in the paper so the first one is quite simple um uh you can't increase colgoma of complexity by conditioning on more States because the program can simply choose to ignore the input if it doesn't help to shorten the length of the program um so we have trivially that KS given M the complexity of conscious experience given the message is at least as great as K as given M and P Phi um but you would expect this Gap to be quite significant uh because of the complexity of um P5 um so because in in our case and generally in non-toy cases um the probability distribution itself provides a lot of information so it it it it significantly reduces the descriptive complexity of s if you're able to um conditional notes this bus results illustrates uh essentially the difference in ineffability from the tabular rasa case on the left where you don't condition on um Alice's brain parameters Phi um uh uh to the case where you can assume access to Alice's uh brain parameters which is analogous to the Shannon entropy uh characterization that that Eric was talking about but the quantity that we're more interested in um is uh this expression in the green box so K as given M and P tilde Phi which is the ineffability expected um well in expectation um of alistair's experience s given the message and Bob's brain parameters uh tildify um you can show that this quantity is up a bounded um by an expression that scales with k p Phi given tilde Phi um which is a measure of the cognitive dissimilarity or the descriptive complexity of Alice's um brain given Bob's brain um so in other words if your experiences are ineffable to someone I.E the the left hand term in the green box is is High um that implies that your brains are highly behaviorally dissimilar under our model um so we can use this this result to provide an account for what Mary learned when she stepped out of her black and white room uh and essentially um uh what it's saying is that neurotypical Alice's experience of color is ineffable to marry um uh which implies that they are cognitively dissimilar and cognitive dissimilarity is not the same as knowledge inadequacy because um knowing how the brain should respond to a particular stimulus is not the same as being able to execute that response when you're actually exposed to the stimulus um so the cognitive dissimilarity um principle is something that we generally believe intuitively but it's also been studied um in Neuroscience uh so the ability for the neural neural activity of two brains to synchronize which is to say to behave in a mutually determined manner appears to facilitate communication between individuals uh there's also a connection to theory of mind which is the skill of being able to infer the thoughts of others so if Bob's cognitive functions um uh FX and Fs which produce his working memory trajectory and his conscious experience are optimized for decoding um Alice's message M into her conscious experience s then ineffability is reduced if um uh Bob's conscious experience as tilde is conditioned on compared to um um because uh it implies that part of the computation of reconstructing um s is executed during inference of tilde s um meaning that the smallest program from uh tilde s Bob's conscious experience um until defy Bob's cognitive parameters to um s Alice's conscious experience would make use of um tilde s to reduce the sort of residual work that that needs to be um uh uh the residual information that needs to be supplied in order to determine s which would shorten the descriptive length of that program so in other words making progress at inferring the conscious experience of others um in your own cognitive processing could literally be interpreted as reducing the ineffability of their conscious experience quickly uh we're going to touch on the grounding problem um so as Eric mentioned before two individuals will generally understand the same word or sentence in different ways and our measure of ineffability does capture this um dissonance for for two reasons um and first uh measures of ineffability um such as we saw uh in the previous slide um they uh they are bound by sealing that scales with cognitive dissimilarity um or the colgoma of complexity of P5 given P tilde Phi um and uh Phi includes all the parameters in Alice's computational graph including those that parameterize functions are defined on the input data D and likewise for Bob so that that means that uh uh Alice's is uh parameters Phi are are grounded in a representation that is at least partly shared with Bob's tildify so if Bob's parameters Implement a function that operates differently on input data um compared to Alice then they do not inform on each other to a great uh extent and the ceiling on ineffability is is increased via this um KP uh Phi given P tilde Phi tan and secondly um conscious experience s uh is a function of Phi it's computed um uh using a function that depends on Phi um and likewise for Bob um and and Phi contains Alice's long-term knowledge therefore um s is capable of containing uh information about the associations that that Alice makes in the process of generating her her conscious experience um and and that's included in the Reconstruction targets of um k s uh which is to say um the description of complexity of her um so our model offers an interpretation for um uh the observations of spelling which remained in 1960 in this very famous experiment where he showed subjects a grid of characters um briefly and then ask them to recall a specific row um he observed that uh subjects were generally able to report the um prompted row accurately uh but um not all the characters in The Grid uh despite being able to report that they had a conscious experience of um they sort of had had they consciously apprehended all uh characters in The Grid um and um our our model uh uh offers an account for this this phenomenon in in the following way so upon being exposed to the grid of characters and being prompted to report the characters in a specific row um working memory uh contents represented by the attractive State a um presumably contains the identities of those four characters in The prompted row as well as a summary over the grid for example the approximate number of characters or their Arrangement and an estimate of the information lost uh by that that summary whilst information is sufficient to discriminate all characters um would exist at some point in the computational pipeline um but uh primarily in Upstream sensory State uh v um from which uh working memory trajectory acts and um working memory output a are computed um subsequently since the attractive State a is directly accessible to to verbal reporting processes um the characters of the prompted Row The Grid details at summary level and the presence of information loss would be directly reportable whereas full grid details uh wouldn't um and and note that this uh explanation would hold irrespective of whether you um where the distinction between conscious and unconscious is drawn um so whether X the working memory trajectory which may contain sufficient information uh to discriminate all characters or not um whether X is considered conscious or not um so as this uh suggests one of the points that we make is that at least when it comes to characterizing richness and ineffability the exact definition of um what constitutes conscious experience so the definition of FS Phi um in the um does it is is not that important we utilize this minimal computational model um without relying on the Implement implementation details in order to allow us to to to make General statements about richness and affability um and as Eric hinted before um you can account for the report of ineffable ineffable uh the report of um uh from this computational assuming this computational graph irrespective of how you define uh FS um so if x is considered to be conscious then the attractor Dynamics bottleneck the amount of information accessible to working memory output and verbal reports but if X is not conscious then information lost during processing can still be reported on it can still be approximately computed and reported on leading us to reports on ineffability um so finally uh some some future directions for this work uh one of them is to uh link it to the Neuroscience um so for example to uh actually get some empirical estimates of the amount of information loss um using neural correlates of working memory state for example um and uh also I think for a deeper understanding of Consciousness you have to consider why it exists and not just how it manifests so this is something that we don't touch on on the paper um but uh the use of information bottlenecks in machine learning would suggest that information loss actually has a purpose and specifically the purpose is generalization um it's it's improves the robustness of functions learned from data uh to noise um and it improves the ability of of functions to perform um obviously optimally on inputs that were not seen during training which is what what generalization is defined by and uh this is important because if we want to incorporate um sort of observations about Consciousness into our artificial models what we really want to capture is the benefit that Consciousness affords biological models um and we might not need to transfer the actual um form that it takes in um human cognition and that's it awesome thank you for the presentation it would be awesome if you could give some opening remarks and then we can have a discussion and hear any questions from the live chat sure well thanks again for the invitation to uh mind jazz on these beautiful topics so as opening questions and discourse I would say like uh a strong message here is that uh through uh the formalization that was just presented we have um an account that allowed to to make the bridge between access Consciousness and phonological Consciousness and uh something that was described very well in in the paper that was cited uh of Jonah nakash in philosophical transaction B how you can you don't necessarily need something more than access Consciousness to a campar is phenomenal aspects but it also let a lot of open questions like typically uh we have seen how there is an information loss within a brain and between brains but we can also complexify uh the the Witten brain uh message passage for instance adding metacognition so at media we are also working on uh architecture on top of global neural workspace uh with typically attention schema Theory from Michael Graziano and so there is a again in this meta or presentational steps a lot of information as well and I think from empirical uh point of view at both behavioral and neurophysiological level there's also a lot to unpack there on how our metacognitive uh uh representation of ourselves is also an impoverished information from the the real uh states that occur and underlying and that connected well with the the social Dimension uh of the of Consciousness and here we touch a bit on that with the unifiability the interpersonal level but in the case of the attention schema there is a very interesting uh reversal from evolutionary standpoint of why the brain comes up to to represent others at first and then we are recycling the the neural mechanisms to predict others behavior on ourselves and like there's a kind of uh flipping of the traditional narrative in cognition Neuroscience with Graziano because the others come first in a way and self-consciousness become a sort of a side effect of having to have all the mechanisms to deal with others and um you know we don't talk too much about the representation of the cell but I think it's a very uh interesting uh pass uh following this work then there's also a something that we didn't discuss too much but uh we we think about the trajectory to the attractor and how the many trajectories can lead to one attractor but interestingly from a sequence of attractor you can recover sort of a trajectory so like if I have a point a and point B and point C by having the sequence IBC I will tend to have a trajectory that looks alike each time and so uh it's kind of heading towards like the Dynamics also of social interaction and culture uh so typically cultural adaptive facts like musics or movies or eliciting stuff from a subjective experience that seems like uh Stronger Than Just Words and uh and even I mean I'm saying stronger than words but even words are sequences so in a way there is maybe in the Dynamics of communication a little bit of information that is retrieved by the interpolation of different discrete states of different attractors so that's uh it's in the uh the inequality that we saw so the inequalities are are there but maybe that we can play a bit and uh with the those phenomena to see to what extent we can retrieve more information by playing on those dynamical phenomena uh in and regarding communication so we talked about the grounding problem and how the the message in the attractors are kind of aligned between people through the shared statistical environment uh and I think the the conograph framework is interesting uh in the sense of a generative model of the of the world that you have to um to share with others and in a sense uh I'm working a lot on autism and I'm interested in neurodiversity uh the the fact that you have this this similarity the cognitive dissimilarity as uh in the uh in the information loss uh it's also very interesting uh to interpret our diversity of the biological level can uh have uh an impact on how certain people will have more difficulty to communicate with others in specifically in a neurotypical world like we have a society that is very normative in in design for neurotypical and so people who are more outside the the norm and are more nor diverse would then have more difficulty to align the genetic model and communicate with others that's also something that would be nice to discuss or to explore later and finally so uh since we are on the on the active inference uh live stream uh I would take the the occasion to to mention or it could uh connect with with active in France we didn't really uh uh talked about the active inference uh formalism in the paper but uh typically I'm coming back from a workshop on computational neurofhinology uh where we we used uh the the previous work of Varela on neuroscience and phonology to try to have a generative passage between first person and third-person experience and we discussed during the whole week our computational model models can be a third uh constraint to to stabilize this uh relationship between person and third person perspective unconsciousness and I was among the rare people there uh not completely convinced by active inference but uh and I I was like trying to argue about uh the need for plurality at the formalism level I mean even from a theoretical perspective and for plurality in those domain but uh at least from a formal point of view to to try to not bet all our eggs in the same uh basket so to say and here it's a very nice example how even you can combine formalism that were mutually exclusive in the literature so for typically uh coming more from the embodied cognition and various work uh people with embodied cognition where tends to be more in dynamic or a System Theory and very opposed to computationalism and information Theory here we show in this paper that information Theory and the energy in theory can totally work hand in hand so it's it doesn't mean that we need to commit to one or two of all formalism but to maintain this priority I think it's very important from epistemological standpoint and still the the connection with active inference that could be interesting and that's one of the things that I kept from last week Workshop is how from a social interaction point of view uh synchronization and a workload on on intervene synchronization and those fundamental synchronization can uh be optimality for information transfer as we see here but in the the context of active inference it's also the optimal uh minimizer of free energy because you have the two generative models that are perfectly aligned so you don't have error in your prediction because the other person become a mirror of your own gnrt model so that may be also the some bridges that and predictions that can show that those different formalisms are multiple way of looking at the same thing so yeah thank you um awesome points so feel free to discuss if you have any responses and just take it from there Eric and shoe or I can ask some questions or read some comments do you have any comments on games thoughts um yeah I think we have some do you have things to say um yeah um that was a long list of comments I I find your comments about the autism um I'll link really interesting um because actually that's that's pretty much directly what the message of the result is is that um cognitive dissimilarity is associated with difficulty in communicating so yeah actually the last thing that that you said about how um if the communicator and the listeners brings are uh synchronized or if they're equal there's no prediction error that that that that that's kind of the machine learning problem actually as a whole um I don't know if this is a big thing but the whole point of machine learning is to discover the true model um that you're trying to learn so that you have no prediction error right so yeah I mean um so I'm not an expert on on active inference by any means uh but one thing that does make me a bit like apprehensive about it is that it's so concerned with prediction you know the the the it's generative models are about reconstruction you know and and um whereas cognition is about survival and prediction is part of survival obviously I mean you need to be able to predict you know I don't know uh not not crashing into a car or something so that you don't die but but it's it's actually you know it's it's it's it's just a part um and more generally uh the inference of cognitive states in the brain is not about discovery of Truth in some sense you know it's not about uh coming up with with with uh true uh ex model or explanation for some phenomenon it's it's about sort of optimizing for your for your Fitness so yeah that's something that that um I don't quite understand uh sort of what the active inference perspective on that would be I don't know if Daniel or you had some some or not okay just jump in with active inference and then I'll leave more comments in the chat but I think that's a great set of points so if I can make one point to guillaume's First you spoke about in the presentation the generalized ineffability and it reminded me of a few times in active inference where we're hearing about generalized synchrony generalized which means like not necessarily lockstep or mirror but mutually information encoding and not necessarily linearly correlate and and synchronized like the end state of the metronomes but there's like complex information transfer especially in this multi-scale setting um generalized synchrony which is enabled by the general generalized coordinates which are the coordinates of motion of the path so these are kind of the summary statistics to describe the Taylor series expansion around that path or to better describe that path taken in some way and generalized homeostasis which for certain environmental regularities it may be sufficient to be entirely retroductive or there may be physical or cyber physical constraints so that you'll have a system that is purely designed a different way so I wouldn't say any theory is necessarily too opinionated with respect to what you could describe and I think part of the reason to want such a descriptive approach which is what I feel like you all did by deciding to describe an analytics framework for ineffability rather than for example a mechanistic explanation but describing some evidence all of that was just to say pluralism in what we're approaching that ineffable space from multiple compressions or bulls to even describe English words scientific models and it reminds me of the recent discussions with Lance DeCosta who described how other discrepancy measures other than free energy could be used and that free energy has been used in machine learning the variational autoencoders all kinds of Bayesian applications because it has a KL diversions and so it has some convenient variational optimization properties as a discrepancy measure but it is not the only discrepancy measure and you mentioned to in your discussion which were the Shannon syntactic sort of classical information content and then the colomogorov the program and so that is like another distance measure in some ways these spaces can't or won't or don't even need to collapse below pluralism of at least several relatively stabilized kinds but I thought that was a very important Point Guillaume about explanatory pluralism and how we can be excited about one area of science or one way of knowing and it's compatible with others ways of knowing because we have some of those properties of differences that were discussed so those are first thoughts before even any octave technicality go for it sure no I just remember Mark that was I mean KL is distance between two distributions whereas uh yeah but anyway yeah get on continue yeah I was just saying like for in the case of the autism uh this typically show our also you don't necessarily need to go to the mechanistic nor biology to have potential explanation and so in the in the in the case of crabber for explaining the brain like there's a very nice definition very nice landscape of different kind of explanation and uh in the same way like those different formalism can give different perspective on how to understand what's going on uh autism has been very focused on the neurobiology at some point and the genetics but maybe it's not at that level specifically that the functional understanding should occur I don't I don't say it has a no cause are relationship because it's highly genetically irritable and so on but uh sometimes the the functional understanding is not necessarily at the uh very uh uh lowest level of uh explanation here so now to kind of move to active inference and really physics based approaches the trajectory recovery is going to have some Maximum information representation with the generalized coordinates of motion it's how paths are represented in many settings and so that path of least action the Bayesian mechanics on that landscape whatever it is empirically it's like fitting a spline in that space or doing some other kind of modeling whether it's SPM modeling or hyper scanning there's some kind of generalized time series modeling between the present and artifacts the past the future other entities what what do you mean Daniel I didn't get it just that that kind of trajectory recovery is what is being parameterized in physics of Consciousness type models or physics of cognition Bayesian physics anything where a probability distribution has a position velocity acceleration and so on it's being path inferred continuous time generative models and there's discrete time generative models in active and Hybrid models just like other Bayesian graphs so in the continuous time setting where there's a Taylor series expansion that's trajectory recovery yeah well I think more generally um maybe the point is that you could use um Bayesian Bayesian learning variational inference to learn a generative model of these states that we're referring to right if you had some kind of data sets uh that actually represents a sample from let's say the true model um then you could use variational inference to to to um to approximate that model but you don't have to though I mean there are many generative modeling techniques in in machine learning um that don't fall under variational inference or read a comment in a question so yahshua benjio writes Guillaume Dumas idea about trajectories of consecutive attractors to recover information about likely trajectories that lead into each of these attractors that are normally lost in working memory is very cool and then Shu Eric any relation between what you talked about and the compositional properties of conscious thoughts yeah I I think so so in terms of compositionality of conscious thoughts I think what yahshua is referring to now is that um our thoughts kind of compose Concepts that we already have so something like uh you know a sentence if I think uh you know the Fat Cat I'm composing these these mental Concepts to fat and Cat and that generates sort of a a new experience so experiences are always generated from these these building blocks um and then similarly one thing that we didn't we didn't talk so much about conscious experiences language and attractors can also have this shared compositional structure so like language it's clear you know face face where it's combined to form new little phrases new sentences that have a novel meaning in a systematic way um how this could happen with the tractors is well kind of in the most basic case different dimensions in this High dimensional State space might converge to different attractors and so now you know one attractor that's converged to is really a composition of attractors along different dimensions um so this is a sense in which you know our experiences are compositional uh our tractors that generate those experiences um can have compositional structure and this could be a reason why language itself is compositional we're basically trying to come up with some some way of describing or identifying compositions of attractors I mean I I would I would ask that compositionality is a mechanism for reducing description complexity um you know if in fact that that's arguably the objective for the compositional nature of our thoughts and language um uh you know features like less time to describe more robustness to noise yeah I mean if you're trying to describe um an instance of a state um obviously the description complexity is much less if it's built up of Concepts that you you already know about yeah just just to bring this back to something I mentioned in the talk um you know I I pointed out that attractors have this discrete structure there's a finite number of them and we could assign labels to them uh but if these attractors are compositional which would allow for richer space Richard number of attractors basically it's finite but it's the space of total attractors is exponentially large right so we don't want a language that's assigning one unique word to every composition of attractors it would be an exponentially large language um we want some kind of uh language that will minimize the description like with what shoe said you know you could have a language that's compositional I think that provides a really I think you're muted Daniel sorry it provides a very interesting and justific justifying rationale for using the description length as opposed to the KL Divergence which is maybe in some telepathy enabled world or space the KL Divergence does help you move quickly or on some defined manifold but in an encoding decoding space and on the computers that we have then writing shorter programs and or making smaller information theoretic compressions becomes one of the imperatives to take the kind of empirical approach that you are describing because though you've described mainly equations these are all also computer packages that can be used to model different data sets so what kinds of data sets do you think this is most proximally applicable understand the question yeah are you are you saying like what kind of data sets would um kind of minimum description length the compositional models be most useful for or just with the approaches and measures that you described today what kinds of data sets or behavioral and cognitive settings do you think that could be used as an empirical tool or measure for I think it'd be really interesting to like hear about does the data already exist is it some hypothetical measurement are there ways to even re-analyze yeah skill or anyone else yeah uh I think like uh on language already there is like a lot of things that have been done on uh the number of uh bytes per second transferred for instance and uh I guess like on large Corpus of discussions uh or even in clinical settings for instance I'm working on uh the the clinical alliance between a patient and a clinician for instance this formalism of unifiability and information loss would be interesting to apply to as even a form of biomarker I mean even if at the language level is not bio and then you can also apply those uh information theoretic measurements to uh physiological data movement uh neural data so in my lab we are recording multiple brains simultaneously we can totally try to Ampere Achilles approach what we are describing this paper uh and and show that for instance in certain population that are more challenged to communicate through in our diversity so what we discussed earlier uh we can apply the same type of measurements with information Theory instead of just a classical uh no Imaging measurements so that's that would be for instance a empirical uh way of dealing with that what I'm very interested in would be to make the relationship between the access Consciousness and the V basically a process and the information loss that you have in your working memory right now the I feel that the no imaging technology are not at that level of uh possibility but who knows maybe we're gonna have a soon the ability to disentangle that kind of stuff yeah oh yes yeah so yeah so the delimitation of what constitutes working memory um in the brain I think that's that's not a a settled question right um I mean there are like lots of different possible approaches yeah I mean it's totally up in the air like uh classically in global workspace Theory they traditionally thought okay working memory is localized to cortex now the the New Perspective is emerging that okay maybe working memory is sort of distributed uh across the brain we see kind of sustained attractor representations uh all over the place in this distributed working memory so all these different pieces um you know what what does what constitutes x what constitutes that working memory trajectory um and and also you know what constitutes The Experience what's the function that goes from X from working memory to an experience uh all that is is up in the air when writing this paper we had plenty of discussions about you know what is the trajectory the whole trajectory of conscious is just the attractor conscious as their kind of sense in which both these things are true so like Guillaume was talking about access versus phenomenal Consciousness uh access Consciousness refers to you know what I'm what I'm able to report whereas phenomenal Consciousness kind of refers more to the the standard definition of you know what is my experience like um so our framework uh kind of illustrates that well maybe we can unify these things where X is a trajectory is some phenomenal experience it's it's in the moment whereas your attractor is really the only thing you're able to report um remember and you some must still know that you know something something is missing between what you're reporting and what you were actually experiencing throughout yeah I mean I think the I mean in in terms of our model is pretty clear um but I think um actually actually grounding it in empirical uh data sets is is is a challenging question the other thing is that cognore of complexity is more of a theoretical measure um so that that's actually I think the reason why Shannon information is much more ubiquitous um uh for example in machine learning uh because basically given the generative model you can you can approximate it um for example just using Monte Carlo um estimation but complexity yeah it would be much more of an approximation with a much greater um error I think if you you attempted to to approximate that so I guess that would be a potential concern yeah one one advantage of it if you if you could measure the columnarok complexity is it would allow you to kind of measure the um the ineffability of an experience on an individual uh case uh so for instance like one one thing we spoke about with theory of Mind was that this complexity of the experience given the message uh we would think it would be much higher than an experience given you know Bob's experience given the listeners experience because the listener's brain is doing a lot of the work of decoding the message into a similar sort of experience so it'd be useful if you could see something like are there certain types of experiences where the message just wildly and insufficient you know the the comograph complexity of the experience given the message is massive uh but actually people do a really good job inferring The Experience um and maybe it would be different in a case of like autism maybe there's some sorts of experiences that they simply can't reconstruct so I think there's some value in trying to create measures of comograph complexity to kind of handle these individual cases the issue is is yeah it's not really computable you can't search through the space of all possible programs but maybe there are kind of ways to get around this for instance you could sort of parameterize a program with I don't know a neural network let's say you have neural network going from message to some some brain scan uh for an experience and then maybe the size of that neural network or the complexity of the network could be um sort of the a surrogate for uh the complexity of the program the shortest program yeah that's a cool idea that is really cool it reminds me of using adversarial neural networks which don't have as strong analytical guarantees as for example cryptography but then they kind of converge in being able to protect or change or modify information in certain ways but you aren't getting the same bounds or guarantees on the formality and that's kind of like the discrete continuous dialectic and so it's important to embody even that pluralism with respect to State spaces because of the formalisms in in any of these domains the formalism for discrete in continuous they're not always aligned and those are definitely the areas that can be studied but for the space of empirical models that we're talking about as part of even then methodological pluralism and science but if we're talking about statistical models we want to be able to intake the data at some point so it has to have certain properties which are even looser than the math and there's probably a lot that could fit in this and I think it'll be interesting like when you have a hyperscanning experiment how do you include the environments or how do you deal with different number of sensors or different types of sensors different kinds of action and report that people might encode their symbols through symbol all kinds of accessibility and interface types which I think returns to the album's point so yes please yeah actually that's also uh remind me that there is also a nice uh empirical playground and see with what we discussed is how we can maximize from environmental perspective instead of the biological perspective this information transfer so I'm working for instance with Michael lip sheets which part of Arius as well on from anthropological standpoint humans came out with uh rituals or cultural practice to to maximize uh the alignments of people and those things also are uh very important in our society or to reach consensus and collaborate and so on and uh I guess like this can also of uh issue would be a very important I mean when we have to uh convert on optimal Collective Dynamics here we are talking with Alice and Bob in a diadic scale but a nice follow-up also on this work will be at the group level how how you optimize consensus and collaboration on on group skills particularly for attacking climate change or social inequality and very challenging topics of society um yeah so actually that touches on uh a well possibly weakness of of the simple model that we use which is that we don't incorporate any feedback loops so we have this very simple um unidirectional model which I think has advantages uh as well because it it sort of shows basic principles in a very in a relatively easy to understand way um but you're right in that uh realistically there's always feedback not just at the interpersonal level but obviously within the brain you know working memory is is top down um attention actually so that's that's something that we didn't address in this in this yeah it would be really interesting to sort of unify our work with the the field of pragmatics which is exactly about this of like you know given many back and forth uh steps in dialogue how do people come to a consensus you know and and this could also uh kind of involve the active inference framework where you know as a listener I'm going to ask questions that are going to try and reduce my uh my uncertainty about the speaker's State as much as possible awesome thank you for these comments so in my experience computer scientists and those who have empirical data measurement experience they know that you can have different kinds of Behavioral measurements or just columns in a database you could have heart rate and you can have video data and you can kind of have this like open-ended it doesn't need to be just a matrix that gets multiplied in this extremely clean way and I feel like that leads a lot of these very open-ended and flexible approaches which on one hand address the different forms and functions and settings that these kinds of questions are asked and like at the same time are working if not to address at least to have continuity with classical discussions on a wide range of topics like sequence yeah the the question of you know not everything is a vector not everything is a matrix you could have more complicated data structures um that's really interesting I think relevance to Consciousness so it certainly seems like our thoughts uh our experiences have this sort of symbolic structure in a way right you can kind of think of them as like any thought sort of a graph I have a couple Concepts in mind maybe in working memory and and I have some idea of how they they relate to each other you know if it's not a graph it's it's something like that maybe some kind of smooth representation of a graph um on the audio on the other hand when you just look at the brain all you see is neurons all you see is this distributed high-dimensional neural activity so there's this really interesting question of of uh how how you link them together and uh I think attractor Dynamics are another uh interesting Avenue for for getting Eric and you could you kind of get a lot of things you get these these high dimensional continuous states that you could represent um you get sort of discrete states that you could represent and they could compose together so um yeah I I I think there's there's a there's something there to kind of extend that point to this very meta science area of discussion around pluralism in science that Guillaume open the the um box on when practice in the behavioral lab if not much more broadly is using structures that have different column types than certain kinds of decisions that might have been taken to be in principle or like Arguments for the superiority even contextually of one framework those are just unveiled purely methodologically as modeler degrees of freedom like whether you use a generative model that's continuous time or discrete time continuous State space discrete State space the thermometer data how you discretized it where you put the thermometer all these broader factors that kind of instantiate the experiment in the pipeline they're just all embodied and cultured Niche events in a scientific niche in a social niche and that's a very pragmatic or people might use other words or like realist or I don't know what adjective would describe that take on starting with the scientific community and formal modeling and the modeling process as its own behavioral object and then working so have the empirical practice more General like leading the generality open-endedness with practice though that raises so many second order questions as well yeah well we can have also from a very pragmatic standpoint like we need to have our producibility so to have like those uh Norms in design of experiment and formal uh tools makes sense but we need to just remind ourselves often that they also constrain our way of thinking and typically uh uh the the story of science shows our mathematics uh is Guided by the problems that we have to solve uh and typically if we take a quantum mechanics uh it was not like the formalism was popped out at first and then uh Quantum physicists use it it's like it was a generative process and a collaborative effort between mathematician and physicists and interestingly they come up even with two different formalism that were shown to be compatible later on so it's a nice tales in a way showing how you can have even a stronger industrial objective consensus by adopting different formalism that then later on are shown to be uh compatible and and equivalent and here I I have unfortunately to to live at some point to pick up my daughter but uh we can go into category Theory and more meta mathematics to to to try to show that those four majors them in the end are potentially uh equivalent but from a history of science I think it's not just about the equivalences also to have different viewpoints that confirm each other uh to make a uh our idea our model of the world uh interceptively stable so to say I'll give you a small short closing comic yum and then we'll talk a little bit more soon Eric um when I first saw some of your figures for hyperscanning and there were nodes that were being connected with different edges that were within and among brains and there was this one part slash dimension of my experience that was like you can't do that they're not connected it's like but it's a statistical causal graph so it they are exactly the same type of explanation in empirical data structure it's the same exact Mutual information linear causation whatever you want to do in edges and Edge in that representation and that's the map territory situation they're not the ones that are touching and it was just like by seeing two levels and the way that the representation could be used it was approved by example what it meant at one level well it was hard to uh to publish at first because I think of that uh reason thank you very much thanks again for the invitation take care so what direction would you like to go or I can read another comment or anyone else in the live chat in our last minute okay let's um kind of uh read last comment from Joshua wrote but these attractors presumably correspond to different concepts in the same sentence are not independently sampled sorry it's not like verbatim but what are these structures in language in terms of like I thought that was very memorable um thing to say that words are sequences it's like they're modular with respect to the dictionary but then even the phonemes or the the writing structures have their own compositional logic like a periodic table and then there's sub logics and then there's just different uses of different levels of description which is why we don't speak the same language unless we do yeah well directly I think what Josh was uh referring to is like that when you when you sample a sentence I guess like clearly these are their own discreet things but you're not going to be sampling them independently from each other it's like different words we'll go here more or less and also they're they'll cohere more or less given a context right you want to form some kind of sentence that describes some some contextures and observations so uh yeah in regards to these these compositional attractors again just a simple picture where you have different attractors converging along different dimensions um obviously these These attractors are going to affect the convergence of others right so it's not like everything is just moving independently along different dimensions um yeah yeah I mean um simplifying assumption that we make in a simple computational model used in the paper um is that we hide uh sort of dependencies as noise um in the variables to avoid uh needing to consider these these links explicitly um to sort of highlight the main um points about richness and affability that that we wanted to make but I really liked your point um Daniel about how there is uh sort of the data there's the physical reality and then there's the compositional structure that is sort of imposed on that I mean it's it's a model right so it's a modeling assumption that you make um and you have to choose at what level of abstraction um to to Define that model and it can feel a bit arbitrary yeah and then you you also you also discussed kind of uh yeah well we all speak different different languages um yeah an interesting question for that in in our work is well okay was it would that imply that we have different attractor structures across different cultures that have different languages is there an interaction there um or are these just you know these different languages equivalent ways of described in the same state of attractors and I think you know it's it's it's somewhere in the middle uh regardless of what language we'll I think similar Concepts in similar uh but there's there's some kind of uh cases where you know you have a word in a given language that uh maybe identifies sort of a fundamental attractor and that's kind of like they're part of their their conscious vocabulary I guess whereas you know we regardless of what language we speak we'd be able to you know get in that conscious State ourselves but maybe given our language like the attractors that you would need to compose to to construct this conscious state would would be a bit more complicated you know you'd need like something like a longer a longer sentence a longer thoughts uh to to have the same sort of experience awesome well one thing that brought to mind was neural patches on the skin like different parts of the body have different density of sensory types from very fine scale touch and actuation to some more broader patches like where two needles you can't determine the difference between them and then that relates to many areas to kind of development of taste and differentiability being able to to determine differences and then you of course use this very like evocative but also grounded dual representation of the samples that are discrete and may have compositional structure with a path that's moving through them and seeking to embed the topology of the sample points within a information geometry which is the kind that our actual statistics input and then the neural patches are regions where to one person it's like yeah I drove through New Mexico and to the other person it's like every inch they're having this Rich experience they know the different stops and then I think one last grade point from the presentation was um that we could kind of I'm not exactly sure I've said but emulate this informational characteristic and maybe not even have that the embodiment for that to even have consciousness of course that'll be the debate then but the debate used to be much less sophisticated than that and also people took principal stances I wasn't totally clear on that that last point that you uh that you made principled stances on whether this would be emulating or simulating or approximating analytical framework for or whether this by implementing that machine that it would be implementing Consciousness versus like the kind of linear regression package that does cognitive modeling that could be understood in a purely SPM like framework and SPM doesn't take a package perspective except maybe implicitly on like whether SPM models generate consciousness most likely it's easy to say no because of generalized linear models so some people may take an in-principible stance that any description or simulation on in certain ways it's all good we're definitely not generating anything conscious it's just would purely be a hugely energetically expensive potentially statistics exercise which is fine yeah so I mean I mean the the the chain of reasoning of all work um I suppose is that if you assume um a certain model of Consciousness then that implies ineffability as it is understood by information loss but obviously information loss does not apply it does not imply consciousness um even though it's an attribute um of of how Consciousness manifests um in humans yeah I mean this this question of of what is the necessary this is a rata to to declare whether a system is conscious or not is sort of very it is not you know what we discuss in the paper at all I mean from my perspective as as a machine learning practitioner um it's it's it's closer to what I said in the last slide I would like to get the benefits of consciousness in in an artificial model and it it doesn't matter so much whether the form that takes um is similar to how it manifests in in biological um systems or not why should it you know I mean we don't have the kind of resources that um Evolution has had over over you know a billion years to to optimize support for this for this model um so the question is you know how do we design how do we Design Systems that that can benefit from for example the generalization and robustness properties um that being very contractive in processing can give you a rather than you know what's the exact definition for Consciousness and how do we replicate that that inner machines because that to me is sort of a very superficial um uh well it's a it's a more superficial problem in that it's looking at the how and not the why yeah yeah I have some thoughts on this also so I mean um Jonathan Simon who's who's a co-author uh on on our work um he gave a really interesting presentation recently there which is you know whatever model of Consciousness you assume let's say Global workspace Theory um there's a minimal model of it that you could construct right like we we could easily construct some kind of uh model like to make this really concrete there's a bunch of mini neural networks that are the sub processes and they're they're communicating uh through some like shared RNN let's say so you could construct the minimal model like this train on some tasks and then you know if if your your Viewpoint was that a global workspace Theory with such and such properties um is what generates Consciousness well then you'd be forced to say that your your minimal model um is conscious and this is in a way I guess not problematic but but sort of counter-intuitive because the model could be doing really simple things they could not even be reporting that it's having any Consciousness at all it might not even have language it might just be doing something like solving mnists in a global workspace type architecture but do you really want to say that that thing is uh is is conscious um so uh yeah it is a a problem I think with uh most series of Consciousness uh maybe one kind of uh exception in my view is theories that say you know we basically have a representation that we're conscious so um one canonical example I guess is uh Michael Graziano's attention schema um which basically says um our brain constructs a model of uh what attention is is doing in the brain and that model basically just describes something that is like experience it describes oh I'm aware of these things uh the they have they have these these properties uh you know I could report that I'm experiencing these things it's basically a representation a set of neural activity uh in in the brain and there um you don't have the same problem where you can construct a minimal model because presumably this representation of Consciousness is uh is really complex um and then I guess another set of of uh theories uh would say that even if you emulate you know whatever processes in the brain we think generate Consciousness um the physical implementation might be really important over here so the example that comes to mind is IIT which would say you know it's basically uh substrate dependent in the sense that you could have the exact same function the exact same mechanism implemented in in different ways like uh you know on a on a computer processor or a neuromorphic hardware and in one case there there may or may not be the Consciousness whereas you have the exact same function in another substrate and yeah it is conscious so um that's that's uh I guess uh another class of theories that would say that emulation does not necessarily imply that you're actually generating uh Consciousness yeah all in our last two minutes what are your closing thoughts or next directions exhortations to the humans and language models yeah well I I guess my um I'm interested in in this stuff a lot from uh purely philosophical uh perspective so um you know the hard problem for me is really Salient of you know why would any physical mechanisms generate Consciousness um and the nice thing at least for for me about this workbook the the most satisfying part of it was that you know I initially had this intuition that experiences you know what the color red is it's not just information it's kind of uh you know under determined there's something more than just information because I can't do things like describe it uh but you know now now we have a theory that kind of uh kind of breaks that that intuition and now I have a satisfying explanation I feel for you know why I can't describe my experiences uh but but under a physicalist framework so I think it'd be interesting to to reason through some uh some other of the the thought experiments that make the heart problem Salient and kind of um yeah to develop some some some models uh that kind of break the intuitions for for the hard problem uh for in my case at least like you know I've seen it done in in this scenario uh with the topic of ineffability so there's no reason in principle that it can't be done uh across other aspects of the hard problem yeah so I think if I were to sum up the work in one very high level sentence it's that um well reasoning subjectively about subjectivity is very difficult but reasoning objective objectively about subjectivity is easier and that's sort of what we do in the paper because um we take an objective standpoint and we try to um you know formalize or characterize uh subjectivity from that standpoint uh since yeah that that sort of makes makes the picture clearer I mean from my perspective I'm interested in Consciousness mainly in terms of what it offers what it brings um uh in terms of for example improving generalization of artificial um land models yeah it's it's interesting because uh so another work that I'm doing is is is um formalizing generalization bounds um for for information bottleneck um which is a regularization principle um used in machine learning so it's just really interesting how evolution has has sort of discovered by itself this this um uh regularizer principle um that that that it it's uh applies to to human cognition foreign actually does improve um guarantees on on generalization um that that is really mind-blowing to me yeah um yeah wow great presentation so thank you both for joining you're welcome back anytime yeah thanks so much Daniel it was a lot of fun are you okay